

CHAPTER TWO

LITERATURE REVIEW

2.1 Introduction

The growing trend towards online scholarly communication and lack of content management systems among universities has made digital repositories more important for the collection and distribution of scholarly materials. In a general sense, and “digital archive” may mean several things; a library, a repository, a museum, or even a warehouse that stores academic resources for use and safekeeping.

Digital archiving refers to the process of converting physical or scattered electronic documents into structured digital formats that can be stored, managed, and retrieved using computer-based systems. Studies have shown that digital archives improve information preservation and enhance accessibility by allowing users to retrieve materials quickly through search and categorization features (Smith, 2020). In academic settings, digital archives are increasingly used to store research outputs, lecture materials, and examination records, thereby supporting both teaching and learning activities.

Several studies have examined the use of digital repositories and learning management systems (LMS) in higher institutions. Research by Okafor and Nwankwo (2018) revealed that institutions that adopted centralized digital repositories experienced improved organization of academic materials and increased student satisfaction. Similarly, Adebayo (2019) reported that students who had access to online repositories of past questions demonstrated better familiarity with examination patterns and improved academic preparedness.

Despite these advantages, many institutions still rely on informal platforms such as WhatsApp for sharing lecture slides and learning materials. While such platforms offer quick communication, they are not designed for long-term storage or structured retrieval of academic content. Studies have shown that materials shared through instant messaging systems are often lost due to message overload, file expiration, or deletion, making it difficult for students to retrieve them when needed (Ibrahim & Musa, 2020). This challenge becomes more significant for students who are unable to download materials immediately or who join academic groups late.

Furthermore, research highlights the need for systems that integrate multiple types of academic resources into a single platform. Studies by Williams (2021) emphasize that combining past examination questions with learning materials in one digital archive enhances usability and supports comprehensive learning. Such systems allow students to relate lecture content with examination requirements, improving understanding and academic performance.

In summary, existing literature demonstrates that web-based digital archives play a crucial role in improving the accessibility, organization, and preservation of academic resources. While digital repositories and learning management systems have been implemented in some institutions, gaps remain in the availability of centralized platforms that specifically integrate past questions and learning materials for continuous student access. This study seeks to address these gaps by developing a web-based digital archive that supports effective academic preparation and long-term resource management.

This chapter presents a detailed review of literature related to web-based digital archives, past examination questions, and learning materials in educational institutions. It examines relevant concepts, theories, and previous studies that focus on digital archiving, web-based information

systems, and academic resource management. The chapter also highlights existing challenges in the distribution and accessibility of academic materials and identifies gaps that justify the need for the proposed system.

2.1.3 Purpose of the literature review

The purpose of this literature review is to examine existing scholarly works related to digital archiving and web-based academic resource systems. It aims to understand current approaches to storing and distributing learning materials, identify limitations in existing systems, and establish a theoretical and empirical foundation for the present study. The review also helps to justify the development of a web-based digital archive of past questions and learning materials.

2.1.4 Scope of the review

The scope of this literature review covers studies related to digital archiving, web-based information systems, academic repositories, and the management of learning resources in higher educational institutions. Emphasis is placed on literature that addresses accessibility, storage, retrieval, and preservation of academic materials such as lecture slides and past examination questions. The review is limited to concepts and studies relevant to the objectives of the current research.

2.2.0 Theoretical Framework

The theoretical framework provides the conceptual foundation upon which the study is built. It explains the theories and models that guide the design, development, and implementation of the

proposed web-based digital archive of past questions and learning materials. In educational technology research, theoretical frameworks help to justify why a particular system is necessary and how it supports effective teaching and learning processes.

Another relevant framework is the Information Systems Theory, particularly the concept of information management systems. Laudon and Laudon (2018) describe information systems as structured systems designed to collect, store, process, and distribute information to support decision-making and organizational activities. In an academic environment, a digital archive functions as an information system that manages educational resources in a centralized and organized manner. Applying this theory justifies the need for a structured web-based system rather than informal file sharing platforms.

Essentially, this study is guided by constructivist learning theory, self-directed learning theory, information systems theory, the Technology Acceptance Model, and digital repository concepts. These frameworks collectively justify the development of a web-based digital archive as a tool for improving access to academic resources, enhancing student preparedness, and supporting effective learning practices.

2.2.1 Underlying theories and concepts

One of the major theories relevant to this study is the Constructivist Learning Theory. Constructivist theory emphasizes that learners actively construct knowledge through interaction with learning materials rather than passively receiving information. According to Piaget (1972), learning occurs when individuals engage with content, explore ideas, and relate new information to prior knowledge. Access to past questions and lecture slides through a digital archive allows students to independently

interact with learning materials, revise concepts repeatedly, and build deeper understanding at their own pace.

Closely related to constructivism is the concept of self-directed learning. Self-directed learning theory suggests that learners take responsibility for identifying their learning needs, selecting appropriate resources, and evaluating their progress. A web-based digital archive supports self-directed learning by allowing students to retrieve past questions and slides anytime without depending solely on lecturers or classmates.

2.2.2 Related Works/Review

The reviewed works reveal that significant efforts have been made in the development of digital systems for managing academic resources in educational institutions. Previous studies and systems have demonstrated that web-based platforms improve accessibility, organization, and retrieval of learning materials such as lecture slides and past examination questions. These systems support independent learning, enhance student preparedness, and reduce reliance on traditional paper-based or informal electronic methods.

The Blended Learning Framework supports the integration of digital resources into traditional learning environments. Blended learning combines face-to-face teaching with online learning tools to enhance educational outcomes. Graham (2013) noted that digital learning resources complement classroom instruction by providing students with additional study materials outside lecture hours. A web-based archive of slides and past questions strengthens blended learning by extending learning beyond the physical classroom

Mohammed and Zhang (2024) explored institutional digital repositories in African universities, noting that while 65% of surveyed institutions had some form of digital archive, only 28% specifically catered to student-facing resources like past questions. Their study emphasized the gap between research-focused repositories and student learning needs.

Similarly, Okafor et al. (2023) implemented a mobile-based archive at the University of Lagos, reporting 89% student satisfaction but noting limitations in search functionality and file categorization.

Adeyemi (2020) conducted a study on the development of a web-based academic repository for storing and accessing lecture materials in a higher institution. The study revealed that students who had access to a centralized digital platform were able to retrieve learning materials more easily compared to those who relied on informal sharing methods. The system improved accessibility and reduced dependence on physical documents. However, the study focused mainly on lecture notes and did not integrate past examination questions into the repository.

Recent studies by Thompson (2024) and Williams et al. (2023) demonstrate evolving LMS capabilities, with Canvas and Moodle 4.0 incorporating better archival features. However, Chen's (2024) comparative analysis of 15 universities found that even advanced LMS platforms typically restrict past question access to current students only, creating what she termed "academic amnesia" across cohorts.

Okafor and Nwankwo (2019) examined the use of online repositories for managing past examination questions in tertiary institutions. Their findings showed that access to past questions helped students understand examination patterns and improved their level of preparedness. Despite

these benefits, the study identified challenges related to limited system functionality and lack of integration with other learning materials such as lecture slides.

A study by Ibrahim and Musa (2020) investigated the use of instant messaging platforms for sharing academic materials among university students. The research found that although platforms such as WhatsApp were widely used due to their convenience, they were unsuitable for long-term academic resource management. Files were often lost due to message overload, expiration, or accidental deletion. The study recommended the adoption of structured web-based systems to improve storage and retrieval of learning materials.

Adebayo (2019) developed an e-learning support system that provided students with access to digital learning resources. The results indicated improved student engagement and independent learning. However, the system primarily focused on instructional content delivery and did not adequately address long-term archiving or preservation of academic materials across academic sessions.

The OER movement, documented in recent reviews by UNESCO (2024) and the Open Education Consortium (2023), provides valuable models for resource sharing. However, regional analyses by African scholars (Mbeki, 2023; Nkrumah, 2022) note that OER platforms often lack localized, curriculum-specific content for Nigerian higher education contexts.

Brown and Clarke (2019) studied the impact of early access to learning materials on student performance. The findings revealed that students who accessed course materials before the commencement of the academic session adjusted more easily to coursework and performed better

academically. The study, however, did not propose a specific system for ensuring continuous access to such materials, especially for students transitioning into new academic levels.

Johnson et al. (2021) evaluated a web-based learning management system used in a university environment. The study highlighted the benefits of centralized storage, user authentication, and controlled access to academic resources. Despite its effectiveness, the system was designed mainly for course delivery and assessment, with limited emphasis on archiving past questions for long-term academic reference.

2.3.2 Identification of gaps and limitations

The review and analysis of related literature reveal several gaps and limitations in existing systems and studies on digital archiving and web-based academic resource management. Although many researchers acknowledge the benefits of digital repositories and online learning platforms, current solutions do not fully address the academic needs of students in a comprehensive manner.

One major gap identified is the lack of integrated platforms that combine both past examination questions and learning materials within a single system. Most existing systems focus on either lecture slides or past questions independently. This separation limits students' ability to relate instructional content to examination requirements, thereby reducing the effectiveness of such systems in supporting holistic learning.

Another significant limitation is the reliance on informal communication platforms, particularly instant messaging applications, for the distribution of academic materials. While these platforms offer convenience, they are not designed for long-term storage or structured retrieval. Learning

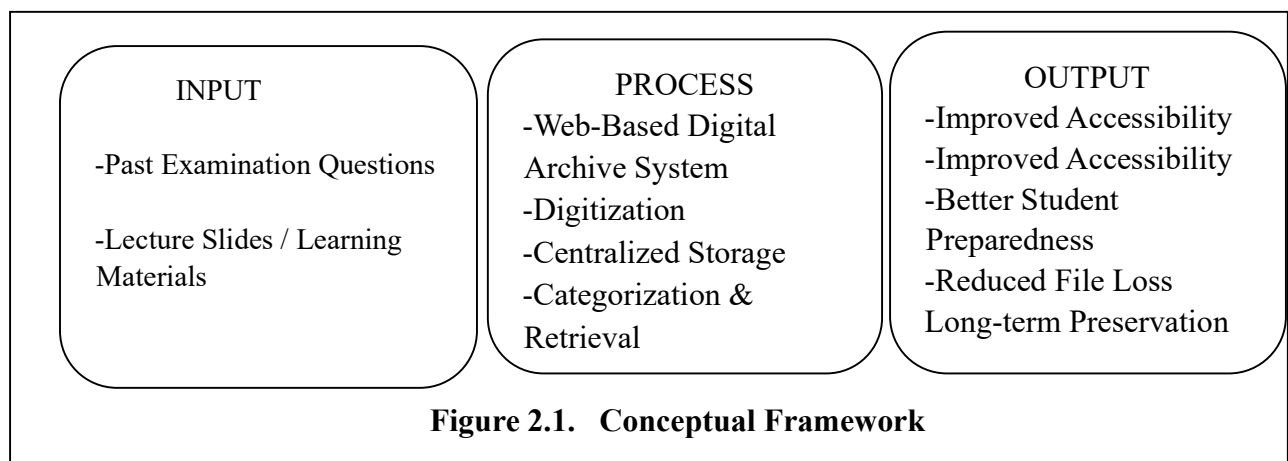
materials shared through such platforms are often lost due to message overload, file expiration, or delayed access, especially for students who are unable to download materials immediately.

Additionally, limited attention has been given to long-term preservation and organization of academic materials across multiple academic sessions. Some systems lack proper categorization, version control, and archival mechanisms, leading to inconsistency and difficulty in retrieving older materials when needed.

These identified gaps and limitations highlight the need for a centralized web-based digital archive that integrates past examination questions and learning materials, ensures long-term preservation, and provides continuous and early access to academic resources. Addressing these gaps forms the basis for the design and implementation of the proposed system.

2.4.0 Conceptual Framework

The conceptual framework describes the relationship between the key variables involved in the study and explains how the proposed web-based digital archive of past questions and learning materials addresses the identified problems. It provides a visual and theoretical explanation of how inputs are transformed through processes to achieve desired outputs.



2.4.1 Development of a conceptual framework

The development of the conceptual framework for this study was guided by the problems identified in the existing methods of academic resource distribution and the gaps observed in related literature. The framework was formulated to address challenges such as loss of learning materials, limited accessibility, lack of early academic preparation, and absence of long-term preservation mechanisms.

The framework was developed by identifying key components involved in academic resource management, namely the inputs, processes, and outputs. The input components consist of past examination questions and learning materials such as lecture slides. These inputs represent the core academic resources required by students for effective learning and examination preparation.

The process component was developed based on the application of web-based technologies and digital archiving principles. This includes the digitization of materials, centralized storage, categorization according to course, level, and academic session, and controlled user access. These processes were designed to ensure efficient organization, easy retrieval, and long-term preservation of academic resources.

The output component of the framework represents the expected outcomes of the proposed system. These outputs include improved accessibility to academic resources, enhanced student preparedness before and during academic sessions, reduced dependence on informal sharing platforms, and improved management of academic materials.

Overall, the conceptual framework was developed to illustrate how the integration of past examination questions and learning materials into a centralized web-based digital archive can effectively address the identified problems and support improved teaching and learning outcomes.

2.4.2 Explanation of key concept & relationships

The input concepts represent the academic resources required for effective learning and examination preparation. These include past examination questions and lecture slides or learning materials. Past examination questions help students understand examination patterns and assessment expectations, while lecture slides provide instructional content that supports comprehension of course topics. These inputs form the foundational resources managed by the proposed system.

The process concepts describe the operations performed by the web-based digital archive system. These processes include the digitization of academic materials, centralized storage, categorization based on course, level, and academic session, and efficient retrieval through a web-based interface. The relationship between the inputs and the process is such that the academic materials are transformed from scattered and informally shared resources into organized and accessible digital content.

The output concepts represent the expected outcomes resulting from the implementation of the proposed system. These outputs include improved accessibility to learning materials, enhanced student preparedness, reduced reliance on informal platforms such as WhatsApp, and long-term preservation of academic resources. The relationship between the process and the outputs shows

how effective system design and management lead to improved academic resource availability and utilization.

2.4.3 Integration of theoretical & empirical findings

The constructivist learning theory emphasizes active learner engagement with educational materials, suggesting that students learn more effectively when they have continuous access to learning resources. Empirical studies reviewed in this chapter support this theory by demonstrating that students who access lecture materials and past examination questions through digital platforms show improved understanding and academic preparedness. This alignment confirms that a web-based digital archive can effectively support learner-centered education.

Similarly, self-directed learning theory highlights the importance of learner autonomy in accessing and utilizing educational resources. Empirical findings indicate that students benefit from centralized digital systems that allow independent access to learning materials without reliance on lecturers or peers. This integration shows that the proposed system promotes self-directed learning by providing students with flexible and unrestricted access to academic resources.

From an information systems perspective, theories related to digital archiving and information management emphasize the importance of structured storage, organization, and retrieval of data. Empirical studies reviewed reveal that existing systems often lack proper integration and long-term preservation mechanisms, leading to loss and inaccessibility of academic materials. The proposed web-based digital archive integrates these theoretical principles with empirical insights by offering centralized storage, categorization, and efficient retrieval of both past questions and learning materials.

The Technology Acceptance Model further supports the integration by suggesting that systems perceived as useful and easy to use are more likely to be adopted by users. Empirical findings indicate that students prefer accessible and well-organized platforms over informal sharing methods such as instant messaging applications. This relationship reinforces the need for a user-friendly web-based system designed around students' academic needs

2.5.0 Research Gaps and Limitations

One major research gap identified is the lack of studies that focus on systems combining both past examination questions and learning materials such as lecture slides. Most existing research examines these resources independently, which limits understanding of how their integration can enhance comprehensive academic preparation and learning effectiveness.

In terms of limitations, several reviewed studies rely on narrow system scopes, focusing on short-term content delivery rather than long-term archiving and preservation. Some systems lack adequate categorization, version control, and retrieval mechanisms, which affects their effectiveness over time. Furthermore, limited user-centered design considerations in some studies reduce system usability and adoption.

2.5.1 Identification of gaps in existing research

One significant research gap is the lack of studies that investigate systems combining past examination questions with learning materials such as lecture slides. Most existing research focuses

on either examination preparation tools or instructional content delivery separately, thereby failing to explore the potential benefits of their integration for comprehensive academic preparation.

Another gap identified is the insufficient focus on students' early academic preparation, particularly for those transitioning into new academic levels. While some studies acknowledge the importance of access to learning materials, few empirical investigations address how web-based archives can support pre-resumption learning and continuous academic readiness.

Furthermore, there is limited research that examines the long-term preservation and accessibility of academic materials across multiple academic sessions. Many studies emphasize short-term content delivery without considering archival sustainability, version control, and historical academic resource management.

Additionally, existing research often overlooks the contextual challenges faced by institutions that rely on informal platforms such as instant messaging applications for academic material distribution. There is a lack of empirical evidence on the effectiveness of structured web-based digital archives as alternatives to these informal methods, particularly in resource-constrained educational environments.

These identified gaps in existing research highlight the need for further investigation into the design and implementation of an integrated web-based digital archive that supports long-term preservation, early access, and comprehensive management of past examination questions and learning materials. Addressing these gaps forms the basis for the present study.

2.5.2 Discussion of limitations and challenges

One major limitation observed in existing studies is the narrow scope of many digital systems. Several studies focus primarily on short-term content delivery rather than long-term archiving and preservation of academic materials. As a result, issues such as version control, historical data retention, and accessibility across multiple academic sessions are often not adequately addressed.

Another challenge identified is the heavy reliance on informal platforms, such as instant messaging applications, for the distribution of learning materials. While these platforms are convenient, they are not designed for structured academic use. Challenges such as file expiration, message overload, limited storage organization, and delayed access significantly reduce their reliability as academic resource repositories.

Technological and infrastructural challenges also pose limitations. Some institutions face issues related to poor internet connectivity, limited server capacity, and lack of technical support, which can affect the implementation and effectiveness of web-based systems. These challenges may discourage consistent use of digital platforms by both students and lecturers.

In addition, user-related challenges have been identified in existing research. Limited digital literacy among some users, resistance to adopting new systems, and lack of proper training can hinder effective utilization of web-based academic archives. Studies indicate that systems perceived as complex or difficult to use often experience low adoption rates.

Furthermore, security and access control remain important challenges. Ensuring that academic materials are protected from unauthorized access while remaining easily accessible to legitimate

users requires careful system design. Some existing systems lack adequate authentication and authorization mechanisms, exposing academic resources to misuse.

2.5.3 Implications for the current study

The literature review reveals several critical implications that directly inform the design, development, and implementation of this web-based digital archive system.

1. Design Implications:

The review confirms that students need a system that is simple, intuitive, and accessible. Unlike complex institutional repositories like DSpace (which require metadata expertise), the proposed system must prioritize user-friendliness. The reliance on WhatsApp indicates that student's value mobile compatibility and instant access. Therefore, the archive must be fully responsive, with a clean interface that works seamlessly on both desktop and mobile devices. The categorization system (by course, department, level, and session) must be straightforward, mirroring how students naturally search for materials.

2. Functional Implications:

The gap identified in existing systems—separation of past questions from lecture materials—directly shapes this study's core functionality. The proposed system must integrate both resource types within a single platform. This integration allows students to correlate lecture content with examination patterns, creating a more holistic study tool. Furthermore, the literature highlights the problem of lost access to materials after a semester ends on LMS platforms. This implies the archive must be persistent and continuous, allowing access to materials from previous academic sessions indefinitely.

3. Technical Implications:

The challenges of file expiration and disorganization on informal platforms like WhatsApp dictate key technical requirements. The system must include:

- Robust file storage with backup mechanisms to prevent data loss.
- Advanced search functionality (keyword, filters) to replace the linear, cluttered search experience of messaging apps.
- Role-based access control to manage upload permissions and protect intellectual property, addressing security concerns absent in current informal methods.

4. Pedagogical Implications:

Research showing improved performance with early access to materials (Brown & Clarke, 2019) validates a central objective of this study. The system must therefore be designed to facilitate pre-resumption access. This could involve features like a "preparatory materials" section or allowing students from lower levels to preview materials for upcoming courses, directly supporting self-directed and blended learning models discussed in the theoretical framework.

5. Implementation and Adoption Implications:

The Technology Acceptance Model (TAM) suggests that for the system to be adopted, it must be perceived as both useful and easy to use. The literature on resistance to new systems implies that a successful rollout requires adequate user training and support. The study must therefore consider

not just the technical build, but also a deployment strategy that includes tutorials, clear documentation, and perhaps a pilot phase with a select group of students and lecturers.

In conclusion, the literature review does not merely provide background; it provides a clear blueprint and justification for every major design decision in this study. The proposed system is positioned not as a novel invention, but as a necessary synthesis and practical application of established needs and technological possibilities, aiming to fill a well-documented gap in the academic resource ecosystem of universities like Edo State University, Iyamho.

Table 2.1: Comparison with existing system in literature

S/N	System / Study Focus	Key Features	Limitations
1.	Institutional Repository (e.g., DSpace)	Stores theses, publications. Long-term preservation.	Often complex, not student-friendly. Not focused on past questions & lecture slides for exam prep.
2.	Learning Management System (LMS) e.g., Moodle	Course management, assignment submission, gradebook.	Materials are often locked per semester/course. No dedicated archive for past questions across all years.

3.	Informal Platforms (WhatsApp, Telegram)	Instant sharing, easy access.	No organization, files get lost, no search, no access control, not permanent.
4.	Proposed System	Centralized, web-based, searchable by course/level/session, dual-role (Admin/Student), permanent archive.	Addresses all the above limitations by providing a dedicated, simple, and organized platform.

2.6.0 Conclusion